

Manthan Sontakke

Nagpur, Maharashtra — +91-8007557800 — manthansontakke005@gmail.com — LinkedIn — GitHub

PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate specializing in embedded firmware development and hardware-software integration, with applied experience in AI/ML-driven signal processing, data visualization and computer vision. Builds IoT and autonomous systems end-to-end from ESP32/Arduino firmware to CNN model deployment.

EDUCATION DETAILS

Vellore Institute of Technology, Bhopal

Aug 2023 – May 2027

VIT UNIVERSITY

B.Tech, Electronics and Communication Engineering (AI & Cybernetics) (Full-time)

CGPA: 8.66/10

TECHNICAL SKILLS

- **Programming Languages & CS Fundamentals:** Python, Computer Networks, SQL, DBMS, AWS (EC2, S3, IAM)
- **Application Development:** FastAPI, Streamlit
- **Embedded Systems & Hardware:** ESP32, Real-Time Systems, RTOS Concepts, MOSFET, BJT, Op-Amps
- **Communication Protocols:** UART, SPI, I2C, PWM, Wi-Fi, Bluetooth/BLE
- **AI/ML & Signal Processing:** TensorFlow, CNN, Digital Signal Processing (DSP), FastAPI, Computer Vision, NumPy
- **Tools & Platforms:** Arduino IDE, MATLAB, LTspice, Streamlit, AirSim, Git, ThingsBoard
- **Professional Strengths:** Technical Communication, Relationship Building, Team Leadership, Persistence & Discipline, Adaptability

PROJECT DETAILS

AI-Powered RF Spectrum Intelligence Platform — GitHub — *TensorFlow, CNN, DSP*

Jul 2026 – Present

- Designed an AI-based platform for automatic RF communication signal generation, analysis, and classification, supporting 5 modulation schemes (BPSK, QPSK, 16-QAM, OFDM, Multi-Tone).
- Architected a DSP pipeline (AWGN noise modeling, low-pass filtering, FFT, spectrogram generation) that reduced manual signal-analysis effort by 80%.
- Trained a CNN on spectrogram-based features achieving over 90% validation accuracy, deployed via an interactive Streamlit dashboard with real-time FFT visualization and confidence-score prediction.

QUENCH - IoT-Based Contactless Smart Water Dispenser — GitHub — *ESP32, Sensors*

Feb 2025 – Apr 2025

- Engineered system/board design for a contactless IoT water dispensing system integrating ESP32, a water flow sensor, managing power design and component integration.
- Programmed embedded firmware to automate dispensing logic and real-time payment verification, reducing manual user intervention by 70%.
- Achieved a dispensing response time under 2 seconds and 95% flow-sensor measurement accuracy for volumes up to 1500 ml through precise sensor calibration and system validation.

Wi-Fi Controlled RC Vehicle — GitHub — *ESP32, PWM*

Sep 2024 – Nov 2024

- Constructed a Wi-Fi controlled RC vehicle for wireless navigation and embedded motor control.
- Achieved response latency below 50 ms using ESP32-based PWM motor control and stable wireless communication.
- Improved movement control reliability by 25% by implementing Finite State Machine (FSM) navigation logic.

INTERNSHIP EXPERIENCE

Embedded System Design Intern

Jun 17, 2026 – Aug 2, 2026

Maven Silicon

- Completed a structured embedded systems course covering digital design and Arduino Uno-based embedded system fundamentals, brushing up on core concepts in microcontroller architecture, GPIO, and peripheral interfacing.
- Built an Arduino Uno-based object detection system for visually impaired users, executing system-level unit and integration tests on sensor input and control logic.

AI/ML Intern

May 20, 2026 – July 31, 2026

MPOnline Limited

- Completed a 6-module curriculum covering Python, data preprocessing, and supervised/unsupervised machine learning, applying feature engineering and data visualization techniques to real-world datasets.
- Trained deep learning models using TensorFlow and PyTorch for Computer Vision and NLP tasks across an 11-week program, and deployed an end-to-end AI capstone project via FastAPI following MLOps practices.

CERTIFICATIONS & EXTRA-CURRICULAR

- **Certifications:** Fundamentals of Digital Design for VLSI Chip Design (L&T EduTech / Coursera)
NPTEL Certified in Internet of Things (Score: 88/100) — Applied Machine Learning in Python (Score: 88.63/100)
- **Extra-Curricular:** Discipline Head, Marathi Cultural Club
Represented at the State Level Cricket Tournament (Under-17, VCA).